# VERA: Support-Aware Certification of Representation Edits Under Deployment Shift

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**Abstract**

A representation edit may appear safe on IID validation data yet damage its target task or expose sensitive information after deployment reweighting. We introduce VERA, a decision layer that asks whether a registered edit should be deployed under a declared shift. VERA jointly audits paired target harm and a fresh heterogeneous attacker portfolio, then returns a simultaneous vector envelope of supported environment- and source-conditional density-ratio budgets, its limiting common radius, or ABSTAIN. We prove finite-sample validity throughout the reported envelope and an impossibility result for new deployment support. Finite-family testing follows Learn Then Test; the proposed contribution is the representation-intervention certificate that couples paired harm, balanced leakage, a continuum of groupwise shifts, and an explicit support boundary. Across 128 preregistered disjoint-seed decisions on four supported vision, text, and time-series datasets, validation-only selection deployed contract-violating edits in 27.3% of cases versus 0.8% for VERA, while VERA retained 51.0% of externally safe opportunities. On Camelyon17, VERA correctly abstained in all 64 registered decisions because the deployment hospital was absent from certification support.

## 1 Introduction

Representation editing is an intervention. It changes what downstream models can recover from a frozen representation, often to remove a protected, nuisance, or source concept. INLP, R-LACE, LEACE, kernel methods, TaCo, SPLINCE, and MANCE++ provide increasingly expressive ways to construct such edits (Ravfogel et al. 2020, 2022a; Belrose et al. 2023; Ravfogel et al. 2022b; Jourdan et al. 2023; Holstege, Ravfogel, and Wouters 2025; Avitan, Goldberg, and Elazar 2026). Their successes do not make every resulting edit deployable. Removing a correlated concept can damage the target, a probe used by the eraser can understate leakage to a different attacker, and an IID validation result need not survive reweighting at deployment.

This paper asks a narrower, operational question: *should this representation edit be deployed under this declared shift?* VERA does not construct a new eraser. It evaluates the output of registered erasers and returns a deployment decision plus a sensitivity certificate: a vector shift envelope, a limiting common radius, the contract that limits that radius, and the support boundary beyond which it must abstain. The target audit is paired. For the same example, it subtracts the identity representation's zero-one loss from the edited representation's loss, isolating incremental harm rather than conflating it with baseline error. The shift class is stratified: within each environment, the target law has a bounded density ratio relative to its certification law. Leakage is the equal mean of two robust source-class recalls, each under a bounded source-conditional density ratio. Environment weights and source prevalence may therefore vary arbitrarily; ordinary prevalence-weighted accuracy is not the contract. At a declared profile, VERA uses a candidate-wise intersection–union test. Separately, it simultaneously upper-bounds every candidate-contract curve and inverts them into a support-aware shift envelope and common radius $\widehat{\Gamma}_e$. A zero radius means that even the IID contract is not established. New support forces abstention.

The statistical mechanism must be positioned carefully. Risk-controlling prediction sets, Learn Then Test (LTT), Conformal Risk Control, and Pareto Testing already calibrate finite algorithm families against one or several risks; recent work also couples risk, acceptance, and utility and provides group-conditional PAC control (Bates et al. 2021; Angelopoulos et al. 2025, 2024; Laufer-Goldshtein et al. 2023; Yu and Liu 2026; Huang et al. 2026). Prompt Risk Control applies related candidate-wise bounds to LLM prompts and includes an estimated covariate-shift correction (Zollo et al. 2024). Weighted risk control and robust validation already address important forms of covariate and distribution shift (Tibshirani et al. 2019; Almeida et al. 2025; Cauchois et al. 2024; Ai and Ren 2024). VERA uses these foundations. Its proposed contribution is the representation-intervention object they do not report: a simultaneous, support-aware envelope of groupwise reweighting budgets for paired edit harm and an entire retrained-attacker portfolio. The common shift radius is one summary of that envelope; unsupported required cells mark the boundary at which it cannot be identified.

Our contributions are threefold. We robustify balanced accuracy into a source-prior-invariant leakage contract and certify its joint support-aware envelope with environment-conditional paired harm over a continuum of deployment budgets. We formalize unsupported-environment abstention by an indistinguishability theorem rather than a heuristic. We then show in a locked, cross-modal official-code study that IID point selection produces unsafe edits that certification rejects, while the full envelope identifies how much declared shift each surviving edit supports.

The stakes are concrete. A hospital may erase acquisition site from pathology features, a hiring system may remove gender from occupational embeddings, a toxicity model may suppress identity mentions, or a gait model may remove recording site. In each case, the edit can reduce measured leakage while damaging the intended prediction or failing in a reweighted group. VERA turns that ambiguity into a falsifiable, support-scoped deployment statement.

## 2 Related Work

**Concept erasure and probing.** Linear erasure includes iterative nullspace projection, adversarial low-rank projection, and the closed-form LEACE solution (Ravfogel et al. 2020, 2022a; Belrose et al. 2023). Kernel erasure, rate-distortion objectives, unaligned-attribute removal, TaCo, SPLINCE, and MANCE extend the construction space (Ravfogel et al. 2022b; Basu Roy Chowdhury et al. 2023; Shao, Ziser, and Cohen 2023; Jourdan et al. 2023; Holstege, Ravfogel, and Wouters 2025; Avitan, Goldberg, and Elazar 2026). Nonlinear guardedness can remain after linear removal, while perfect erasure faces fundamental utility limits (Akbari, Afshari, and Boddeti 2025; Chowdhury et al. 2025). FARE and MMD-B-Fair provide important certificates for restricted fair-representation objectives (Jovanovic et al. 2023; Deka and Sutherland 2023). VERA neither replaces these erasers nor certifies absence of all information. It audits their candidate outputs against a fixed portfolio. That portfolio follows the broader lesson that a probe's result depends on its hypothesis class and training protocol (Hewitt and Liang 2019; Pimentel et al. 2020; Voita and Titov 2020; Elazar et al. 2021).

**Risk control and abstention.** LTT turns calibration into testing over a finite family, and Pareto Testing extends the view to multiple risks and objectives (Angelopoulos et al. 2025; Laufer-Goldshtein et al. 2023). Conformal Risk Control treats general monotone losses (Angelopoulos et al. 2024). Joint adaptive certificates couple risk, acceptance, and utility, while group-conditional PAC methods control routing risk within registered groups (Yu and Liu 2026; Huang et al. 2026). These works invalidate any claim that VERA invents finite-family, joint, or groupwise acceptance. Prompt Risk Control is a particularly close application template: it selects LLM prompts using high-probability bounds on mean, tail, and dispersion risks and corrects covariate shift using unlabeled target data (Zollo et al. 2024). VERA instead audits paired representation interventions without a target sample or estimated importance weights, reports worst-case guarantees over a declared bounded reweighting class, retrains a heterogeneous attacker portfolio, and makes unsupported cells an explicit impossibility boundary. Selective classification and SelectiveNet reject individual examples (El-Yaniv and Wiener 2010; Geifman and El-Yaniv 2017, 2019); VERA instead rejects an entire intervention and provides a predeployment, support-scoped sensitivity certificate.

**Distribution shift and robust evaluation.** Known importance weights can restore conformal or high-probability risk control under a specified covariate shift (Tibshirani et al. 2019; Almeida et al. 2025). Robust validation and fine-grained robust conformal inference provide ambiguity-set guarantees (Cauchois et al. 2024; Ai and Ren 2024), while robust model evaluation and CVaR concentration supply closely related technical tools (Najafi, Sani, and Farnia 2025; Thomas and Learned-Miller 2019; Waudby-Smith and Ramdas 2024). Distributional fairness certificates and Wasserstein fairness audits are close application-level neighbors (Kang et al. 2022; Ehyaei, Farnadi, and Samadi 2025). Worst-case subpopulation treatment effects are especially close to our paired-harm view (Jeong and Namkoong 2020). VERA does not claim these tools or generic robust fairness auditing. It applies bounded reweighting to a selected representation edit's paired target and retrained-attacker contracts and reports the admissible support-scoped envelope. Group DRO and robustness benchmarks motivate the observed-group view (Sagawa et al. 2020; Koh et al. 2021; Sohoni et al. 2020), and domain-adaptation impossibility results motivate explicit support assumptions (David et al. 2010; Subbaswamy, Adams, and Saria 2021).

## 3 Problem Setup

Let $Z \in \mathbb{R}^d$ be a frozen representation, $Y$ its target label, and $S$ a source or sensitive label. An edit $e \in \mathcal{E}$ is fitted without the certification fold and maps $Z$ to $Z^e$. The identity is $e = 0$. Fresh target models are fitted after every edit. For certification example $i$, define paired target harm

$$H_{e,i} = \mathbf{1}\{f_e(Z_i^e) \neq Y_i\} - \mathbf{1}\{f_0(Z_i) \neq Y_i\} \in \{-1, 0, 1\}. \tag{1}$$

In the real protocol, identity is the paired comparator rather than a selectable erasure action. If a deployment protocol may select identity, it must register identity as one additional candidate in the finite-family correction; the separate family-size coverage grid does so. For each preregistered attacker $a \in \mathcal{A}$, let $L_{e,a,i} = \mathbf{1}\{a_e(Z_i^e) = S_i\} \in \{0, 1\}$. Attackers are retrained after each edit using construction data, but are fixed before certification outcomes are used.

For validation law $P$ and $\Gamma \geq 1$, define

$$\begin{aligned} \mathcal{Q}_\Gamma(P) &= \left\{ Q \ll P : 0 \leq \frac{\mathrm{d}Q}{\mathrm{d}P} \leq \Gamma \quad P\text{-a.s.} \right\}, \\ \rho_{\Gamma,P}(V) &= \sup_{Q \in \mathcal{Q}_\Gamma(P)} \mathbb{E}_Q[V]. \end{aligned} \tag{2}$$

The target contract uses $P_g$, the certification law conditional on environment $g$. For binary source, leakage uses the balanced robust accuracy

$$B_{e,a}(\eta_0, \eta_1) = \frac{1}{2} \sum_{s \in \{0,1\}} \rho_{\eta_s, P_s}(L_{e,a}), \tag{3}$$

where $P_s$ is source-conditional. Balanced accuracy itself is standard (Brodersen et al. 2010); our change is to robustify

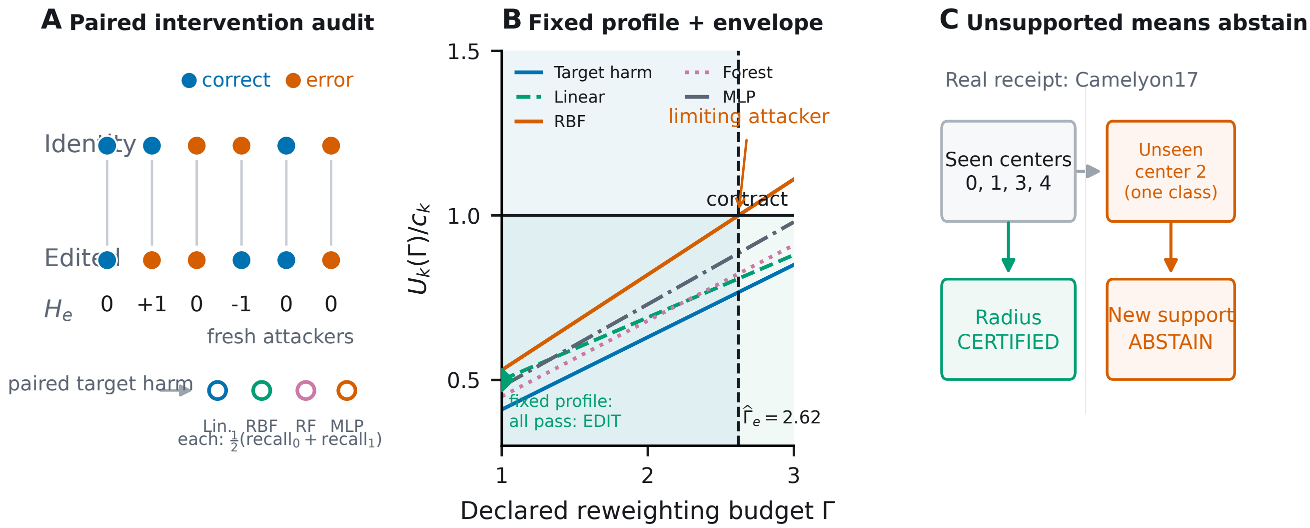


Figure 1: **VERA certifies an edit only within a declared, supported shift class.** (A) Target harm compares identity and edited predictions on the same examples; fresh heterogeneous attackers measure equal-weight source-class recall. (B) At a fixed profile, every component must pass the candidate IUT; simultaneous upper-risk curves are separately inverted to obtain the shift envelope and limiting common radius. Curves are schematic. (C) The guarantee cannot cross new support. In the registered Camelyon17 split, certification observes centers 0, 1, 3, and 4, while external center 2 is absent, so a five-center deployment claim requires abstention unless an additional cross-center assumption is supplied.

each recall and couple the resulting attacker contracts to paired edit harm. A constant attacker scores $1/2$, not one, and changing source prevalence does not change the estimand. At zero or unit prevalence this is a standardized two-conditional functional, not ordinary accuracy observable in an absent class. The population common erasure shift radius is

$$\Gamma_e^\star = \sup\{\Gamma \geq 1 : \rho_{\Gamma,P_k}(V_{e,k}) \leq c_k \quad \text{for every registered contract } k\}, \tag{4}$$

with value zero if a contract fails at $\Gamma = 1$.

This radius is a sensitivity statement, not an estimate of how much shift will occur. A user declares a profile of plausible density-ratio caps; VERA states whether the available certification sample supports the contracts throughout that profile. Conditioning target harm on environment prevents a favorable environment mixture from hiding damage in a smaller group. Standardizing leakage to equal source-class weight similarly prevents source prevalence from making an attacker appear weak. These two conditionings need not describe one factorized joint model: the claim applies when each registered conditional law obeys its own stated cap.

## 4 VERA

The bounded-reweighting risk has the CVaR dual

$$\rho_{\Gamma,P}(V) = \inf_{\eta \in [a,b]} \{\eta + \Gamma \mathbb{E}_P[(V-\eta)_+]\}. \tag{5}$$

For general bounded variables, a Dvoretzky–Kiefer–Wolfowitz band gives a uniform upper curve. Our discrete audit variables admit tighter exact curves. For Bernoulli leakage with success probability $p$, $\rho_{\Gamma,P}(L) = \min(1, \Gamma p)$. For paired harm with probabilities $(p_-, p_0, p_+)$ on $(-1, 0, 1)$, set $a = \Gamma p_+$ and $b = \Gamma(1 - p_-)$. The adversary places its permitted mass on $+1$, then zero, then $-1$:

$$\rho_{\Gamma,P}(H) = \begin{cases} 1, & a \geq 1, \\ a, & a < 1 \leq b, \\ a + b - 1, & b < 1. \end{cases} \tag{6}$$

VERA substitutes one-sided Clopper–Pearson bounds. For paired harm it uses an upper bound $U_+$ for $p_+$ and lower bound $L_-$ for $p_-$, clips $\widetilde{p}_+ = \min(U_+, 1 - L_-)$ to preserve the simplex, and splits the component error between the two tails. Balanced leakage averages two class recall bounds, splitting its component error between the source classes.

At the fixed deployment profile, each candidate is an intersection–union test: all of its target and balanced-leakage components must pass at level $\delta/|\mathcal{E}|$. If a candidate is unsafe, false acceptance implies undercoverage of at least one fixed violated component. A union bound is needed over candidates, but not over components. The separately reported envelope uses Bonferroni coverage over every candidate–contract pair because every curve remains available for post-certification inspection.

Let $U^T_{e,g}(\gamma_g)$ denote the simultaneous target-harm upper curve in observed environment $g$, and let $U^L_{e,a,s}(\eta_s)$ denote attacker $a$'s source-class recall curve. VERA's central reported object is

$$\widehat{\mathcal{E}}_e = \{(\boldsymbol{\gamma}, \boldsymbol{\eta}) : U^T_{e,g}(\gamma_g) \leq \tau \quad \forall g, \quad \tfrac{1}{2}\textstyle\sum_s U^L_{e,a,s}(\eta_s) \leq \lambda \quad \forall a\}. \tag{7}$$

Coordinates for unsupported required cells are zero. The

**Algorithm 1** VERA fixed-profile selection and shift envelope

**Require:** Registered edits $\mathcal{E}$, contracts $c_k$, error $\delta$, budget cap $\Gamma_{\max}$
1: Fit every edit, target model, and leakage attacker without certification outcomes
2: Allocate $\delta/|\mathcal{E}|$ to each candidate's fixed-profile IUT
3: **for** each edit $e$ **do**
4: Accept the profile only if every component bound clears its threshold
5: Separately construct simultaneous curves and envelope $\widehat{\mathcal{E}}_e$
6: **end for**
7: **return** the registered utility-best accepted edit, its envelope, or ABSTAIN

common radius is the largest diagonal point $(\Gamma\mathbf{1}, \Gamma, \Gamma)$ in this envelope, right-censored at the registered computational cap. Reporting the vector envelope as well as its diagonal summary reveals which environment, source class, or attacker limits an edit.

**Theorem 1** (Fixed-profile VERA)**.** *Fix $M$ candidate edits and all audit variables independently of certification. At one declared shift profile, accept a candidate only if every component upper bound clears its threshold at level $\delta/M$. Then the probability of deploying any candidate that violates at least one declared target or balanced leakage contract is at most $\delta$.*

For an unsafe candidate, fix one violated population component. Accepting that candidate implies undercoverage of this component; intersecting more tests cannot increase that probability. Union bounding only over the $M$ candidates proves the result. This is the LTT-style finite-family decision used for the primary experiment.

The distinction from simply applying LTT is in the population contract and output being certified. Each edit induces paired target variables and a family of retrained-attacker variables; each variable is robustified over a continuum of conditional reweightings; and all coordinates are inverted into an admissible shift region. LTT supplies the fixed-family testing logic. The robust edit-specific vector envelope, its limiting common radius and contract, balanced conditional leakage functional, and explicit support boundary define the deployment object studied here. The separation already occurs at the population level: a Bernoulli leakage risk $p = 0.04$ satisfies an IID threshold $0.05$, yet at $\Gamma = 2$ its robust risk is $\min(1, \Gamma p) = 0.08$ and violates the same threshold. An IID LTT decision can validly certify the first statement; it does not imply the second or report the largest supported shift.

The simultaneous envelope is richer but more conservative. It contains every profile $(\boldsymbol{\gamma}, \boldsymbol{\eta})$ for which each environment target curve clears $\tau$ and each attacker's mean of two source-class curves clears $\lambda$. Its coverage event is uniform over the continuum of profile coordinates. Environment mixing preserves the target threshold by linearity, and source prevalence does not enter balanced leakage. Restricting this envelope to the diagonal gives the reported common radius. Full statements and proofs are in the supplement.

**Theorem 2** (Support-aware envelope)**.** *With probability at least $1-\delta$, simultaneously for every registered edit $e$, every profile in $\widehat{\mathcal{E}}_e$ satisfies all target-harm and balanced-leakage contracts. Environment mixture weights and source prevalence may be selected after certification without another multiplicity correction.*

The proof event covers every edit, environment, attacker, source class, and every $\Gamma \geq 1$. Conditional robust-risk bounds imply each component contract; arbitrary mixtures of controlled environment risks remain controlled by linearity, and source prevalence is absent from the standardized leakage functional. Intersecting or choosing objects already covered by the same event does not spend additional error probability. This argument establishes the external-distribution guarantee only for laws inside the declared envelope.

**Theorem 3** (Unsupported-cell impossibility)**.** *Suppose a required audit cell has zero validation probability, and the model class contains safe and unsafe worlds that induce identical protocol observations but disagree on the contract in that cell. If deployment may concentrate there, any validation-only protocol with uniform false-acceptance control $\delta$ accepts the edit in the safe world with probability at most $\delta$.*

The proof is a two-world indistinguishability argument. It applies to an unseen environment or a missing environment–source-label cell; a single-class slice cannot identify the omitted class's conditional leakage. Camelyon17 center 2 and its constant binary source label therefore trigger abstention. This does not assert that the measured center-2 target outcome necessarily violates a contract.

The impossibility result is deliberately stronger than "we have too little data." If safe and unsafe worlds induce exactly the same observations, any randomized protocol accepts with the same probability in both. Uniform false-acceptance control in the unsafe world therefore limits acceptance in the safe world to at most $\delta$. More optimization or a different confidence interval cannot remove this identification boundary; the remedy is new support or an explicit structural assumption.

## 5 Experimental Protocol

An initial five-seed pilot exposed that maximum class recall gives a constant attacker leakage one; those results are retained as an estimand audit and are excluded from confirmation. The balanced study was then fixed in a committed, hashed preregistration before any seed 5–12 run. It crosses five erasers (INLP, R-LACE, LEACE, TaCo, MANCE++), five datasets (Waterbirds, Camelyon17-WILDS, CivilComments-WILDS, Bios, GaitPDB), and eight untouched seeds, totaling 200 official-method runs. The modalities span natural images, medical images, text, and time series. Bios and CivilComments audit known representation-bias settings (De-Arteaga et al. 2019; Borkan et al. 2019); Camelyon17 is a reliability benchmark, not clinical evidence. Each run records split hashes, train-only preprocessing, its upstream commit and entry point, per-example audit-array hashes, and the shared runner commit. Model cards, datasheets, auditing, and reproducibility guidance motivate this receipt design (Mitchell et al. 2019; Gebru et al. 2021; Raji et al. 2020; Pineau et al. 2021).

The official frontier contains 12 edits per dataset–seed block: INLP ranks $\{1, 2, 4, 8\}$, R-LACE ranks $\{1, 4\}$, closed-form LEACE, TaCo removal counts $\{1, 2, 3, 5\}$, and one registered MANCE++ configuration. Every method calls a pinned upstream entry point; no proxy implementation contributes a row. We cap eraser training, construction, certification, and external evaluation at 8,000, 2,000, 8,000, and 8,000 examples. Train-only standardization and, where needed, randomized PCA precede editing. Eraser fitting, post-edit target-probe fitting, and attacker fitting use no certification outcomes.

Each candidate retrains a logistic target probe and four leakage attackers: logistic regression, an RBF Nystroem map followed by logistic regression, a random forest, and a two-layer MLP. Certification uses fractions $\{0.05, 0.10, 0.25, 0.50, 1.00\}$, target thresholds $\tau \in \{0.05, 0.10, 0.20\}$, leakage thresholds $\lambda \in \{0.60, 0.70, 0.80\}$ and $\delta = 0.05$. The IID baseline profile uses $\Gamma = 1.0$ at the full certification size; a fixed $\Gamma = 1.01$ robust sensitivity and the full simultaneous radius are also reported. We compare always deploy, feasible point selection without intervals, IID LTT/VERA-IUT at $\Gamma = 1$, the VERA shift envelope, and an external oracle. The IUT row tests VERA's multicomponent edit contract but is not itself evidence of robustness to $\Gamma > 1$; that evidence comes from the fixed shifted profile and envelope. External contract violation is reported separately from procedural non-certifiability. Camelyon17 external balanced leakage is NA because the slice has one source class, and VERA must abstain because center 2 is absent during certification.

Within each configuration, the registered selection utility minimizes the largest validation balanced leakage, then the largest environment-conditional paired harm, then a stable candidate key. Point selection applies this order to empirically feasible edits without intervals. IID LTT/VERA-IUT applies it only to edits passing the $\Gamma = 1$ fixed-profile certificate. The envelope rule additionally requires the simultaneous common radius to reach the profile. Always-deploy uses the same utility order without a feasibility gate; the external oracle is an evaluation ceiling and never supplies a training or selection label.

We report two endpoints separately. A measured external violation requires an estimable external target and balanced-leakage contract and an actually deployed edit that fails either threshold. A procedurally unsupported deployment enters a registered environment or source-label cell absent from certification. The latter is a failure of identification, not evidence that an observable external threshold was exceeded. This separation prevents the support theorem from manufacturing empirical failures by definition.

The exact balanced study contains 2,000 replicates in each of 54 cells: six certification sizes, three $\delta$ levels, and three shift budgets. Its finite multinomial/binomial law gives an analytic abstention probability before Monte Carlo is observed. A separately hashed 216-cell grid varies certification size, four eraser-family sizes plus identity, validated-environment count, and $\delta$, with 2,000 repetitions per cell. The real learning-curve overlay is labeled as a full-certification plug-in diagnostic, not independent theorem validation. Threshold configurations share eight seed-level fits, so an exact one-sided seed-blocked sign-flip test governs the point-selection comparison under its paired sign-symmetry null, with Holm correction over the four externally estimable datasets. A sign-only test is a lower-power sensitivity that does not require magnitude symmetry.

After that confirmation was frozen, an independent replication locked one stress contract per supported dataset and 32 disjoint seeds (13–44) before execution. Its 800 official-method receipts contain 128 externally estimable deployment decisions plus 64 Camelyon17 decisions across the two VERA rules. Dataset-level point-versus-IUT violations use exact one-sided paired McNemar tests with Holm correction across the four supported datasets. The global VERA false-acceptance rate and safe-retention ratio use exact binomial intervals. The preregistered composite also required point selection to violate at least 20% of decisions in every dataset; this design criterion is reported even when the paired safety comparison succeeds.

# 6 Results

**Independent disjoint-seed replication.** Across 128 preregistered decisions from 32 disjoint replication seeds per supported dataset, point selection deployed contract-violating edits in 35/128 decisions (27.3%), versus 1/128 (0.8%) for IID LTT/VERA-IUT. VERA retained 52/102 external-oracle opportunities (51.0%). Camelyon17 forced abstention in all 64 registered VERA decisions because the deployment hospital remained outside certification support. All four dataset-level paired comparisons favored VERA after Holm correction. The composite preregistered endpoint nevertheless did not pass: 3/4 datasets cleared every condition because GaitPDB's point-selection violation rate was 5/32 (15.6%), below the locked 20% threshold. Dataset-level stress counts, Holm-adjusted paired tests, and the three-panel replication figure are generated as audited artifacts and included in the supplementary archive.

**The certificate pays for safety with abstention.** The aggregate independent result is not a claim that VERA improves prediction accuracy. It shows that uncertainty-aware selection rejects many edits that look feasible at a validation point estimate. Retention rises from 1/146 (0.7%) at 5% of the certification fold to 45/146 (30.8%) at the full fold in the original confirmation matrix, matching the expected evidence–abstention tradeoff. At the prespecified $\Gamma = 1.01$ robust sensitivity, the IID IUT deployment rate fell from 12.5% to 10.3% with 0/288 measured violations. The simultaneous envelope then exposes the limiting environment or attacker rather than reducing robustness to a binary decision.

**The attacker portfolio matters.** In the confirmation matrix, the full four-attacker audit deployed in 12.5% of configurations with zero measured violations. A linear-only audit deployed in 43.6% but violated 98/288 (34.0%) external contracts when judged by the complete portfolio; linear plus RBF

| Dataset | Modality | Target $Y$ | Erased/audited source $S$ | Target environment/support |
|---|---|---|---|---|
| Waterbirds | vision | bird type | background/place | place; supported |
| Camelyon17-WILDS | medical vision | tumor status | center 0 vs. centers 1–4 | hospital; center 2 unseen |
| CivilComments-WILDS | text | toxicity | identity mention | identity mention; supported |
| Bios | text | profession (28-way) | gender label | gender label; supported |
| GaitPDB | time series | Parkinson status | recording site | recording site; supported |

Table 1: Cross-modal protocol and support scope. "Supported" means every registered external environment is represented during certification; it does not imply that the bounded-reweighting assumption is known to hold. Camelyon17 is included to instantiate the unsupported-environment boundary, not to make a clinical claim.

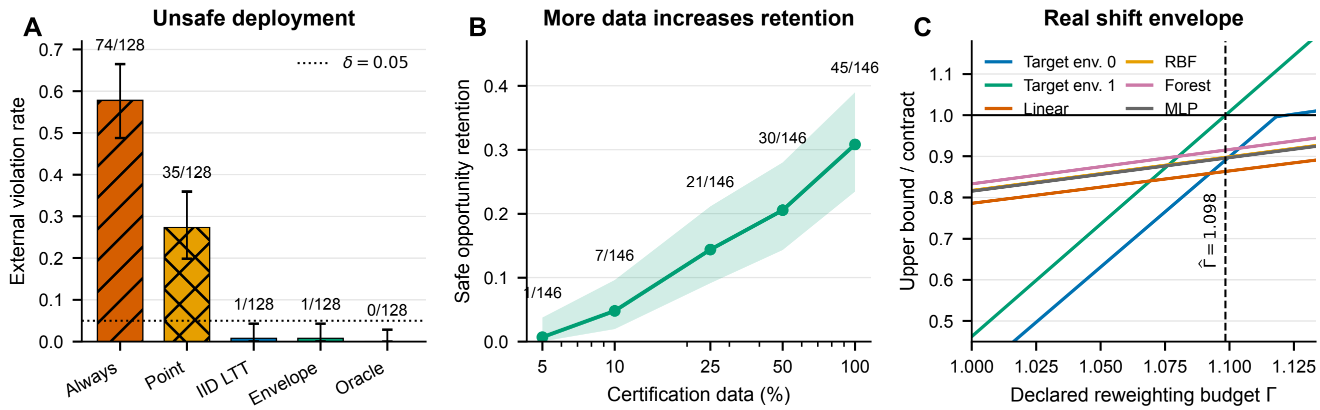


Figure 2: **Certification changes deployment, tightens with evidence, and reports where shift tolerance ends.** (A) Independent 32-seed replication on four supported datasets; bars show contract-violating decisions with exact 95% binomial intervals. (B) Externally safe VERA deployments divided by external-oracle opportunities as certification data increase; this is a full-data plug-in diagnostic, not an independent coverage test. (C) A real CivilComments TaCo receipt shows simultaneous target and attacker upper-risk curves; the first threshold crossing determines the certified common radius.

| Dataset | Point viol. | VERA viol. | Holm $p$ | Safe kept |
|---|---|---|---|---|
| Bios | 12/32 | 1/32 | 0.0020 | 4/7 |
| CivilComments | 10/32 | 0/32 | 0.0029 | 16/32 |
| GaitPDB | 5/32 | 0/32 | 0.0312 | 0/31 |
| Waterbirds | 8/32 | 0/32 | 0.0078 | 32/32 |
| Total | 35/128 | 1/128 | – | 52/102 |

Table 2: Independent 32-seed replication on supported datasets. Violations use all registered decisions as the denominator; Safe kept is externally safe VERA deployments over external-oracle opportunities.

still violated 17.7%. Dropping only the MLP doubled deployment to 25.0% and produced 43/288 violations. Forest, MLP, RBF, and linear attackers were the limiting component in 22, 10, 9, and 4 of the 45 full-portfolio deployments, respectively.

**Theory tracks controlled data.** The exact balanced study had zero false acceptances in 108,000 trials, and all 54 abstention rates fell inside simultaneous exact prediction bands. A second 216-cell grid varied candidate count, validated-group count, certification size, and $\delta$; every cell passed simultaneous coverage. These controlled studies validate the implementation of the finite-sample calculation. The external benchmarks instead diagnose performance under observed shifts and do not prove that those shifts belong to the declared ambiguity set.

## 7 Reproducibility

Code, hashed preregistrations, compact receipts, audit scripts, and manuscript sources are available at https://github.com/RudraChopra/vera-edit-or-abstain. Large public-dataset derivatives remain off GitHub; manifests and content hashes identify them.Every claim-grade result is regenerated from a receipt that records the locked parent commit, upstream eraser commit, split hashes, and per-example audit-array hashes.

The receipt audits validate 1,000 dataset–method–seed runs and zero proxy rows: 200 confirmation runs plus 800 disjoint-seed replication runs. Independent replays rehash raw audit arrays, reconstruct candidate rows, and recompute intervals, eligibility, radii, selected edits, external contract outcomes, and all reported summary statistics. Numeric, selection, and external-semantic mismatches are zero.

**Generative-AI assistance disclosure.** OpenAI Codex was used extensively for research ideation, literature discovery, theorem and proof drafting, software implementation, ex-

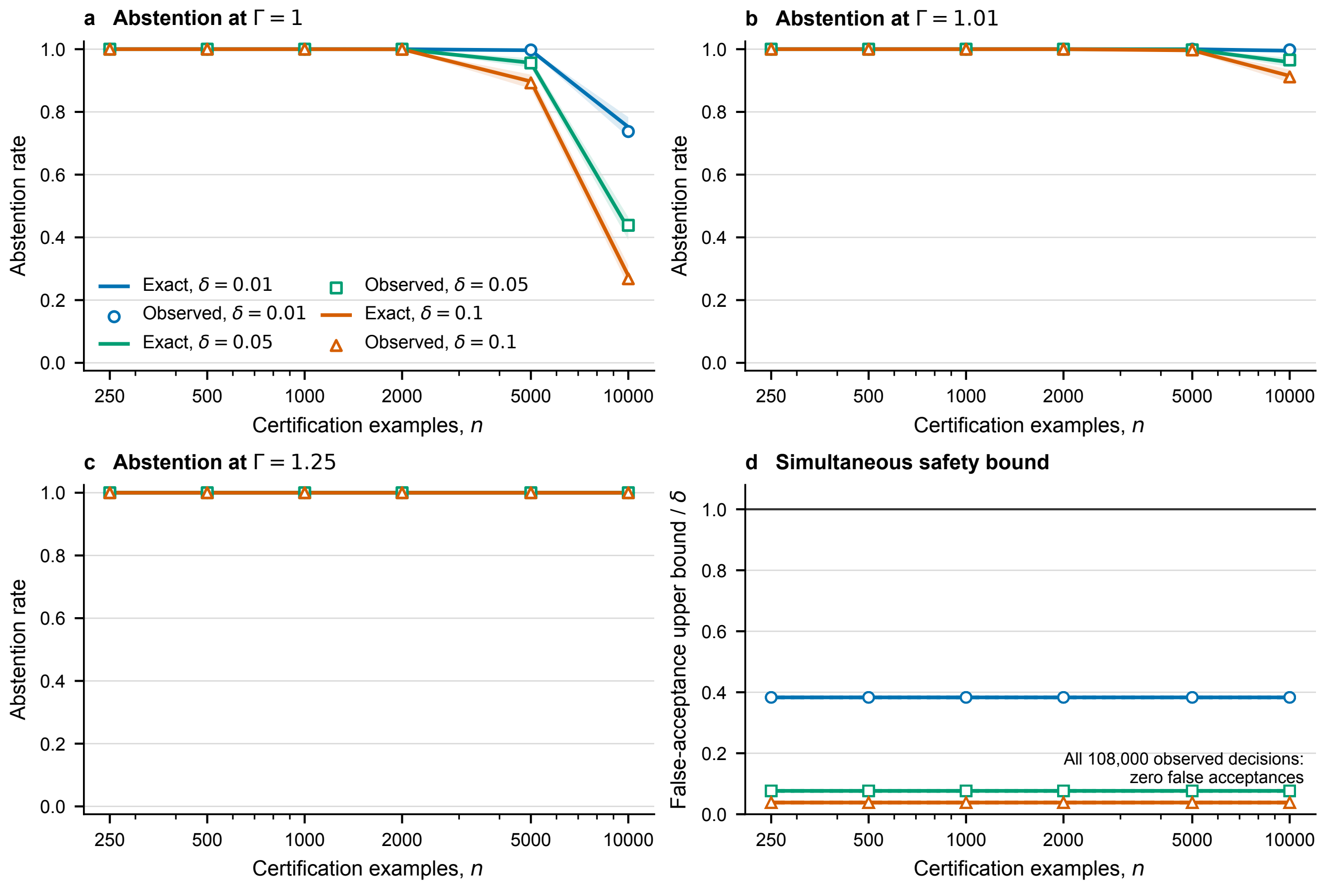


Figure 3: **Predicted and observed abstention coincide in the exact study.** Observed rates from 2,000 repetitions in each of 54 cells overlay the analytic prediction; bands are simultaneous over registered certification sizes.

periment orchestration, statistical-analysis code, figure and manuscript drafting, and editing. No AI system is an author or a citable source. The human author or authors are responsible for independently verifying every claim, proof, implementation, reference, data right, and policy obligation before submission; that verification remains an explicit presubmission gate.

## 8 Limitations and Conclusion

VERA's guarantee is finite-sample but scoped. It assumes bounded audit variables, a certification fold independent of edit and attacker construction, and deployment support covered by certification. The density-ratio budget is a declared sensitivity parameter, not an empirically discovered fact. A finite attacker portfolio cannot prove that no measurable algorithm recovers the concept, and suppressing registered leakage does not establish fairness for unmeasured concepts. The Camelyon17 experiment is not clinical validation. Finally, algorithmic seeds quantify fitting and split variation but do not make the five benchmark datasets a random sample of deployment domains. The original confirmation uses seed-blocked sign-flip and sign-only analyses; the independent replication uses paired McNemar tests over 32 disjoint seeds. Every adjusted value is reported regardless of outcome. Most importantly, an external benchmark violation is evidence about that observed benchmark law, not proof that the law lies inside VERA's declared density-ratio class.

Within those boundaries, VERA changes the output of representation erasure from "apply this edit" to a falsifiable statement: this edit meets these paired target and attacker contracts up to this supported reweighting budget. When the data cannot justify that sentence, abstention is the result rather than a hidden limitation.

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